Training Dynamics

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What you'll learn

Motivation

- Know about some big open questions in learning dynamics
- Understand the concept of implicit regularization
- Know about the different approaches to studying learning dynamics
- Understand the AI safety motivations for learning dynamics
- Know about emergent misalignment as a safety-relevant phenomena that illustrates the importance of understanding generalization for AI safety

Geometry and dynamics of deep neural networks

- Know key results about the loss landscape of deep linear networks (DLNs): critical points are saddles or global minima
- Explain the edge of stability phenomenon
- Understand that gradient flow can be written as NTK-weighted gradient in function space
- DLNs are degenerate and have conserved quantities through gradient flow

Regimes of learning in deep linear networks

- Understand the role of initialization, width and depth for the lazy and rich (saddle to saddle) regimes in DLNs
- Know about implicit regularization from non-linearity (neural race reduction)

References: [SMG14], [AMG24], [TAJ24].

1 Setup and notation

We study deep linear networks (DLNs) with L weight matrices $W_1 \in \mathbb{R}^{d_1\times d_0}, W_2 \in \mathbb{R}^{d_2\times d_1}, \dots, W_L \in \mathbb{R}^{d_L\times d_{L-1}}$. The network computes:

$$f(x) = W_L W_{L-1} \cdots W_1 \, x =: W x$$

where $W = W_L \cdots W_1 \in \mathbb{R}^{d_L\times d_0}$ is the end-to-end (or "student") matrix. We train on a dataset $\{(x_\mu, y_\mu)\}_{\mu=1}^N$ with the squared loss:

$$\mathcal{L}(\theta) = \frac{1}{2N}\sum_{\mu=1}^{N} \|y_\mu - W_L \cdots W_1 \, x_\mu\|^2$$

In the population limit with whitened inputs $\Sigma_X = I$, this becomes:

$$\mathcal{L}(W) = \frac{1}{2}\|M - W\|_F^2$$

where $M = \Sigma_{YX}\Sigma_X^{-1}$ is the "teacher" matrix (the OLS solution) with SVD $M = U \operatorname{diag}(s_1, \dots, s_r, 0, \dots, 0)\, V^\top$, and $s_1 \geq s_2 \geq \cdots \geq s_r > 0$.

2 Loss landscape geometry

This problem explores the critical point structure of deep linear networks, following [AMG24].

Exercise 2.1 (Diagonal decomposition). Consider the **diagonal** case: $d_0 = d_1 = \cdots = d_L = d$, and the teacher is diagonal, $M = \operatorname{diag}(s_1, \dots, s_d)$, with $s_1 > s_2 > \cdots > s_d > 0$. Restrict attention to diagonal weight matrices $W_l = \operatorname{diag}(w_l^{(1)}, \dots, w_l^{(d)})$. Show that the loss decomposes into d independent scalar problems:

$$\mathcal{L} = \frac{1}{2}\sum_{\alpha=1}^{d}\left(s_\alpha - \prod_{l=1}^{L} w_l^{(\alpha)}\right)^2$$

Exercise 2.2 (Scalar critical points). For a single scalar mode with target $s > 0$, find all first-order critical points of $\ell(w_1, w_2) = \frac{1}{2}(s - w_1 w_2)^2$ at depth $L = 2$. Show that the critical points are:

- The global minimum manifold: $w_1 w_2 = s$
- The origin: $w_1 = w_2 = 0$

Classify the origin as a saddle point by computing the Hessian H of ℓ at $(0, 0)$ and showing it has both positive and negative eigenvalues.
Hint: Write the gradient equations

Exercise 2.3 (Critical-point structure). Achour et al. [AMG24] show that every first-order critical point $\theta = (W_1, \dots, W_L)$ satisfies: there exists a subset $S \subseteq \{1, \dots, d\}$ such that

$$W = W_L \cdots W_1 = P_S\, M$$

where $P_S = U_S U_S^\top$ is the orthogonal projector onto the span of the left singular vectors $\{u_\alpha\}_{\alpha \in S}$ of M. For the diagonal case $M = \mathrm{diag}(s_1, \dots, s_d)$, write the value of the loss at the critical point corresponding to $S = \{1, \dots, r\}$ with $r < d$.

Exercise 2.4 (Symmetry of the global minima). The group $GL_h := GL_{d_1} \times \cdots \times GL_{d_{L-1}}$ acts on the weights by:

$$(W_1, \dots, W_L) \mapsto (g_1 W_1,\ g_2 W_2 g_1^{-1},\ \dots,\ W_L g_{L-1}^{-1})$$

Verify that the student map $\mu(\theta) = W_L \cdots W_1$ is invariant under this action: $\mu(g \cdot \theta) = \mu(\theta)$. What does this imply about the dimension of the set of global minima in parameter space?

3 Gradient flow and conserved quantities

We now study gradient flow: $\dot\theta(t) = -\nabla_\theta \mathcal{L}(\theta(t))$, the continuous-time limit of gradient descent with infinitesimal learning rate.

Exercise 3.1 (Gradient-flow equations). For a two-layer diagonal DLN ($L = 2$) with loss $\mathcal{L} = \frac{1}{2}\|M - W_2 W_1\|_F^2$, assume that each weight matrix W_i is a diagonal matrix and whitened inputs. Derive the gradient flow equations for each layer. Show that:

$$\dot W_1 = W_2^\top (M - W_2 W_1), \qquad \dot W_2 = (M - W_2 W_1) W_1^\top$$

Exercise 3.2 (Balancedness is conserved). Define the **balancedness matrix**:

$$G := W_2^\top W_2 - W_1 W_1^\top$$

Show that G is conserved under the gradient flow, i.e. $\dot G = 0$. In other words, gradient flow is constrained on the balanced manifold

$$\mathcal{M}_\beta := \{\theta \in \Omega \mid W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell,\ \ell = 1, \dots, L-1\}.$$

Hint: Compute $\dot G$, substitute the gradient flow equations and verify that the terms cancel pairwise.

From now on, assume **balanced initialization**: $G(0) = 0$, which by Exercise 3.2 means $W_2^\top W_2 = W_1 W_1^\top$ for all time. We want to derive the gradient flow in function space (the ODE for the student $W = W_2 W_1$). This requires several steps.

Exercise 3.3 (Function-space velocity). Compute $\dot{W} := \frac{d}{dt}(W_2 W_1)$ using the gradient flow equations from Exercise 3.1. Show that:

$$\dot{W} = (M - W)\, W_1^\top W_1 + W_2 W_2^\top\, (M - W)$$

Exercise 3.4 ($W_1^\top W_1$ in terms of W). We now need to express $W_1^\top W_1$ and $W_2 W_2^\top$ in terms of $W = W_2 W_1$. For simplicity, assume that the weight matrices W_l are diagonal. Show that:

$$W_1^\top W_1 = (W^\top W)^{1/2}$$

Exercise 3.5 ($W_2 W_2^\top$ in terms of W). Similarly, show that $W_2 W_2^\top = (W W^\top)^{1/2}$.

Exercise 3.6 (The balanced function-space ODE). Substitute the results of Exercises 3.4 and 3.5 into Exercise 3.3 to obtain:

$$\dot{W} = (W W^\top)^{1/2}(M - W) + (M - W)(W^\top W)^{1/2}$$

Exercise 3.7 (The NTK operator). Define the NTK operator for $L = 2$ as:

$$K[F] := (W W^\top)^{1/2}\, F + F\, (W^\top W)^{1/2}$$

Show that the gradient flow from Exercise 3.6 can be written as $\dot{W} = K[M - W]$. It turns out that this result generalizes to depth L on the balanced manifold and to non-diagonal weight matrices:

$$\dot{W} = \sum_{k=1}^{L} (W W^\top)^{\frac{L-k}{L}} (M - W)(W^\top W)^{\frac{k-1}{L}}$$

This is the NTK equation in the case of DLNs. The NTK equation is a gradient flow in function space with NTK being a preconditioning operator for the gradient.

4 Rich regime: saddle-to-saddle

This is the core problem. We derive the exact solution of the rich regime dynamics directly from the self-consistent equation of Section 3, emphasizing the NTK perspective.

Alignment assumption. Start from the balanced gradient flow equation derived in Exercise 3.6:

$$\dot{W} = (WW^\top)^{1/2}(M - W) + (M - W)(W^\top W)^{1/2}$$

We work in the **rich regime** (small initialization) with balanced weights. Assume that the student is **aligned** to the teacher: the singular vectors of $W(t)$ coincide with those of M at all times. Concretely, write:

$$M = U \operatorname{diag}(s_1, \ldots, s_d)\, V^\top, \qquad W(t) = U \operatorname{diag}(w_1(t), \ldots, w_d(t))\, V^\top$$

where U, V are the left/right singular vectors of M, and $w_\alpha(t) \geq 0$ are the evolving singular values of W.

Exercise 4.1 (Aligned NTK). Under this alignment assumption, show that the NTK operator is aligned to the task, i.e. $K[F]$ preserves the SVD basis. In particular, show that:

$$(WW^\top)^{1/2} = U \operatorname{diag}(w_1, \ldots, w_d)\, U^\top, \qquad (W^\top W)^{1/2} = V \operatorname{diag}(w_1, \ldots, w_d)\, V^\top$$

and that the residual is $M - W = U \operatorname{diag}(s_1 - w_1, \ldots, s_d - w_d)\, V^\top$.
Hint: Recall that for $A = U \operatorname{diag}(\sigma_i)\, V^\top$, we have $AA^\top = U \operatorname{diag}(\sigma_i^2)\, U^\top$, and therefore $(AA^\top)^{1/2} = U \operatorname{diag}(|\sigma_i|)\, U^\top$.

Exercise 4.2 (Decoupled scalar ODEs). Substitute into the self-consistent equation. Show that the matrix equation decouples into d independent scalar ODEs:

$$\dot{w}_\alpha = 2\, w_\alpha\, (s_\alpha - w_\alpha), \qquad \alpha = 1, \ldots, d$$

Hint: Compute the product $(WW^\top)^{1/2}(M - W)$ in the SVD basis. It is diagonal with entries $w_\alpha(s_\alpha - w_\alpha)$. The second term contributes identically, giving the factor of 2.

Exercise 4.3 (Timescale of learning). The ODE $\dot{w} = 2w(s - w)$ is a logistic equation whose solution is:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}$$

where $w_0 = w(0)$. Compute the time t_α it takes for mode α to travel from initial strength w_0 to a final strength w_f. Show that:

$$t_\alpha = \frac{1}{2s_\alpha} \ln\left(\frac{w_f\,(s_\alpha - w_0)}{w_0\,(s_\alpha - w_f)}\right)$$

Deduce that modes with larger singular values s_α are learned faster. This is a **separation of timescales**: the network learns features in decreasing order of their singular value strength.

Exercise 4.4 (Incremental learning). Consider a teacher with r nonzero singular values and a small uniform initialization $w_\alpha(0) = w_0 \ll s_r$ for all α. Discuss:

- What is the approximate rank of $W(t)$ at early times?
- How does the rank evolve at intermediate times?
- What happens in the limit $t \to \infty$?

What does this tell us about the sequence of critical points visited by the gradient flow?

5 Lazy regime

We now show that large initialization freezes the NTK and eliminates the timescale separation found in Section 4. For clarity, we work with the **diagonal scalar** model from Exercise 2.1, so each mode α is independent. This avoids matrix algebra and isolates the essential mechanism.

Setup. Consider a single mode: a depth-2 diagonal DLN with scalar weights a, b learning a target $s > 0$. From Sections 3 and 4, the balanced gradient flow for $w = ab$ is:

$$\dot{w} = 2w(s - w)$$

The factor $2w$ is the (scalar) NTK — it is **state-dependent**: the effective learning rate depends on the current value of w.

Exercise 5.1 (Linearizing the NTK). Now initialize at a **large** value $w_0 \gg s > 0$. Assume that in the early phase of training, while w has not changed much from w_0, the ODE becomes approximately:

$$\dot{w} \approx 2w_0(s - w)$$

Exercise 5.2 (Frozen-NTK solution). Solve the linearized ODE from Exercise 5.1. Show that:

$$w(t) = s + (w_0 - s)\, e^{-2w_0 t}$$

What is the learning timescale?

Exercise 5.3 (No timescale separation). Now restore the mode index α. In the lazy regime, each mode satisfies $\dot{w}_\alpha \approx 2w_0(s_\alpha - w_\alpha)$ with the **same** rate $2w_0$ for all α. Compare the lazy learning time t_α^{lazy} with the rich-regime timescale from Exercise 4.3: $t_\alpha^{\text{rich}} \propto 1/s_\alpha$.
Discuss the difference in **rank bias** between the lazy and rich regimes.

6 Mixed dynamics: unifying lazy and rich

In practice, networks are neither purely lazy nor purely rich. Tu, Aranguri & Jacot (2024) provide a unified description. We develop the key ideas using the diagonal scalar model.

Exercise 6.1 (The interpolating ODE). In Sections 4 and 5, we studied the same ODE $\dot{w}_\alpha = 2w_\alpha(s_\alpha - w_\alpha)$ in two limits: small w_0 (rich) and large w_0 (lazy). A more general model for a depth-2 DLN with balanced initialization at scale σ and width n replaces the scalar NTK $2w_\alpha$ by:

$$\dot{w}_\alpha = 2\sqrt{w_\alpha^2 + \tau^2}\,(s_\alpha - w_\alpha)$$

where $\tau := \sigma^2\sqrt{n}$ is a **threshold** parameter that depends on the initialization scale and width.
Verify that this ODE reproduces the two known regimes:

- **Rich limit** ($\tau \to 0$): recover $\dot{w}_\alpha = 2|w_\alpha|(s_\alpha - w_\alpha)$.
- **Lazy limit** ($\tau \to \infty$ with w_α bounded): recover $\dot{w}_\alpha \approx 2\tau(s_\alpha - w_\alpha)$.

Exercise 6.2 (Two-phase dynamics). At initialization, all modes start at $w_\alpha(0) \sim \sigma^2 \ll \tau$ (since $\sqrt{n} \gg 1$). Observe that:

- When $|w_\alpha| \ll \tau$, the effective learning rate is $\approx 2\tau$, the same for all modes (lazy behavior).
- When $|w_\alpha| \gg \tau$, the effective learning rate is $\approx 2|w_\alpha|$, which is mode-dependent (rich behavior).

Describe the resulting two-phase dynamics in words: what happens first, and what happens later?

Exercise 6.3 (Mode-by-mode transition). Not all modes cross the threshold τ at the same time: which modes cross it first? Explain why this means that a single network can simultaneously have some modes in the rich regime and others still in the lazy regime.

7 Stochastic implicit bias (bonus)

SGD introduces noise from mini-batching. Write the SGD update as a combination of gradient and noise term (use a LLM to check your answer). A continuous model is the Langevin SDE:

$$d\theta_t = -\nabla \mathcal{L}(\theta_t)\, dt + \sqrt{\eta\, \Sigma(\theta_t)}\, dB_t$$

where $\Sigma(\theta)$ is the covariance of the stochastic gradient noise and η is the learning rate.

Exercise 7.1 (Boltzmann equilibrium). The probability density $p(\theta, t)$ of the parameters evolves according to the Fokker–Planck equation:

$$\partial_t p = -\nabla \cdot \mathbf{j}, \qquad \mathbf{j} = -\nabla \mathcal{L}(\theta)\, p(\theta) - \frac{\eta}{2} \nabla \cdot (\Sigma(\theta)\, p(\theta))$$

where $\mathbf{j}$ is the probability current. Assume: (i) stationarity $\partial_t p^* = 0$, (ii) thermal equilibrium $\mathbf{j} = 0$, and (iii) isotropic noise $\Sigma = \sigma^2 I$. Show that the equilibrium distribution is the Boltzmann distribution:

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta\sigma^2} \mathcal{L}(\theta)\right)$$

Hint: Setting $\mathbf{j} = 0$ with $\Sigma = \sigma^2 I$ gives $\nabla \mathcal{L}\, p + \frac{\eta\sigma^2}{2} \nabla p = 0$. This is a first-order ODE for p in terms of $\mathcal{L}$. Try the ansatz $p \propto e^{-\beta \mathcal{L}}$ and solve for β.

Exercise 7.2 (Temperature and flatness). The ratio $\frac{2}{\eta\sigma^2}$ plays the role of an inverse temperature β. Interpret what happens to the equilibrium distribution when:

- η is very small (low temperature)
- η is very large (high temperature)

Which regime favors flatter minima, and why might this be beneficial for generalization?

Exercise 7.3 (Anisotropic noise). In practice, SGD noise is **not** isotropic: $\Sigma(\theta)$ depends on both the loss landscape and the data. Without solving anything, explain qualitatively why anisotropic noise can introduce an implicit bias that goes beyond what the loss function $\mathcal{L}$ alone would select. Specifically, why might SGD preferentially escape sharp directions of the loss while remaining stable along flat directions?

Learn more

The readings are roughly ordered in a way that makes sense for learning. This is a curated list of what I find important, and it is not exhaustive.

Key readings

- The loss landscape of deep linear neural networks: a second-order analysis Achour et al. — First-order and second-order classification of critical points in DLN loss landscapes. Classifies strict and non-strict saddles.
- Saxe et al. A mathematical theory of semantic development in deep neural networks — Read the supplementary materials section to understand the exact solution of gradient flow dynamics in deep linear networks
- Geometry of fibers of the multiplication map of deep linear neural networks Simon Pepin Lehalleur et al. — Stratifies global minima into orbits using quiver representation theory
- The geometry of the deep linear network Govind Menon — Beautiful mathematical treatment of gradient flow in DLNs and a surprising mathematical result relating gradient flow and free-energy minimization. Makes implicit regularization explicit (under balanced assumptions)
- Emergent Misalignment is Easy, Narrow Misalignment is Hard — When finetuned on data with harmful inputs, the model generalize in harmful ways on other unrelated datasets.
- Emergent Misalignment is Easy, Narrow Misalignment is Hard — By regularizing, we can mitigate emergent misalignment
- Literature Review: There Will Be a Scientific Theory of Deep Learning
- Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability
- Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Loss landscape geometry

- Deep Learning without Poor Local Minima, Kenji Kawaguchi — For non-bottlenecked DLNs, all local minima are global

- The loss landscape of deep linear neural networks: a second-order analysis Achour et al. — First-order and second-order classification of critical points in DLN loss landscapes. Classifies strict and non-strict saddles.
- Pure and Spurious Critical Points: a Geometric Study of Linear Networks Matthew Trager et al. — Defines spurious minima and counts the number of connected components of the global minima
- Geometry of fibers of the multiplication map of deep linear neural networks Simon Pepin Lehalleur et al. — Stratifies global minima into orbits using quiver representation theory
- The Loss Surfaces of Multilayer Networks Anna Choromanska et al — No-local-minima results in non-linear multilayer networks under some strong assumptions (spin glass models)
- Loss Surfaces, Mode Connectivity, and Fast Ensembling of DNNs, Timur Garipov et al — Study the connectedness of global minima of loss landscapes (mode connectivity)
- The Multilinear Structure of ReLU Networks, Thomas Laurent, James von Brecht — Show that local minima are typically singular
- Classic result: Neural networks and principal component analysis: Learning from examples without local minima Pierre Baldi et al. — Original result that linear network landscapes have no spurious local minima (single hidden layer case)
- Good review of key DLNs results: Gradient Flow Equations for Deep Linear Neural Networks: A Survey from a Network Perspective, Joel Wendin, Claudio Altafini — Also covers gradient flow

Implicit biases of gradient flow

- Saxe et al. A mathematical theory of semantic development in deep neural networks — Read the supplementary materials section to understand the exact solution of gradient flow dynamics in deep linear networks
- Saddle-to-Saddle Dynamics in Deep Linear Networks: Small Initialization Training, Symmetry, and Sparsity Arthur Jacot et al. — Studies the saddle-to-saddle dynamics in DLNs. Introduces the different regimes (lazy and rich)
- The geometry of the deep linear network Govind Menon — Beautiful mathematical treatment of gradient flow in DLNs and a surprising mathematical result relating gradient flow and free-energy minimization. Makes implicit regularization explicit (under balanced assumptions)
- Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures Saxe et al. — Simplicity bias of gradient flow from DLNs extends to non-linear and transformer architectures under small initialization

- Abide by the Law and Follow the Flow: Conservation Laws for Gradient Flows — Gives a recipe to exhaustively derive conservation laws through gradient flow

- SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics — SGD builds up complex solutions by first learning simpler solutions and then composing them

- Mixed Dynamics In Linear Networks: Unifying the Lazy and Active Regimes — Unifies lazy and rich in 2-layer DLNs and gives conditions for the transition between them

- Neural Tangent Kernel: Convergence and Generalization in Neural Networks Arthur Jacot, Franck Gabriel, Clément Hongler — NTK paper that describes gradient flow in function space

- The Neural Race Reduction: Dynamics of Abstraction in Gated Networks Saxe et al. — Implicit biases toward shared representation in non-linear networks as modelled by gated DLNs

- Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks Daniel Kunin et al. — A theory of feature learning as a two-step process between dormant and active neurons

- From Lazy to Rich: Exact Learning Dynamics in Deep Linear Networks — Studies the transition between lazy and rich regimes in DLNs with balancedness parameters

- Implicit Regularization in Matrix Factorization — Gradient descent on matrix factorization converges to the minimum nuclear norm

- Gradient Descent Maximizes the Margin of Homogeneous Neural Networks — Implicit bias toward max-margin in classification

- A Convergence Analysis of Gradient Descent for Deep Linear Neural Networks Sanjeev Arora et al. — Convergence analysis of SGD

Learning rate (discrete GD)

- Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability — Shows that curvature stabilizes around the inverse of the learning rate.

- Understanding Optimization in Deep Learning with Central Flows — GD with a discrete learning rate is equivalent to gradient flow on an effective loss

- Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective — Pretraining exhibits a river valley loss landscape and gives intuition about the learning-rate schedule (warm-up, stable, decay)

- Optimization on multifractal loss landscapes explains a diverse range of geometrical and dynamical properties of deep learning — Models the landscape as multifractal and analyses sub- and super-diffusive behaviour of GD

Stochasticity

- On the implicit regularization of Langevin dynamics with projected noise — Makes explicit the implicit bias of stochasticity in deep linear networks with a Langevin model
- Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning — Golden Path Hypothesis: in the transient regime dominated by drift (typical of pretraining), SGD is simply a noisy deformation of GD (it does not select different basins)
- Stochastic Collapse: How Gradient Noise Attracts SGD Dynamics Towards Simpler Subnetworks — Stochasticity induces a bias toward simpler solutions in deep linear networks
- Stochastic Training is Not Necessary for Generalization — Makes the implicit regularization of stochasticity explicit
- Implicit Bias of SGD for Diagonal Linear Networks: a Provable Benefit of Stochasticity — Shows that stochasticity has a bias toward sparser solutions relative to GD
- Stochastic gradient descent performs variational inference, converges to limit cycles for deep networks — SGD minimizes some free energy and can have circular currents at convergence
- Stochastic Gradient Descent as Approximate Bayesian Inference — Classic paper on SGD locally approximating Bayesian inference, although it assumes non-degeneracies
- A Diffusion Theory For Deep Learning Dynamics: Stochastic Gradient Descent Exponentially Favors Flat Minima — Implicit bias toward flat minima
- Implicit Regularization or Implicit Conditioning? Exact Risk Trajectories of SGD in High Dimensions — Shows the Golden Path Hypothesis on quadratic loss in a convex setup using an interesting SDE model of SGD.
- An Empirical Model of Large-Batch Training — Gradient noise scale can be used to decide the optimal batch size
- Almost Bayesian: The Fractal Dynamics of Stochastic Gradient Descent — Relates SGD to Bayesian training

- Catapults in SGD: spikes in the training loss and their impact on generalization through feature learning — Loss spikes when training with SGD impact generalization
- The Heavy-Tail Phenomenon in SGD — SGD noise can be heavy-tailed, and this induces a different regularization

Emergence: empirical examples

- Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets — Grokking: delayed generalization on modular addition
- In-context Learning and Induction Heads — Induction heads form during training. They are important for in-context learning
- Emergent Misalignment: Narrow finetuning can produce broadly misaligned LLMs — Emergent misalignment: fine-tuning on insecure code can generalize to misaligned behaviour on other data
- Emergent Misalignment is Easy, Narrow Misalignment is Hard — By regularizing, we can mitigate emergent misalignment
- Neural Networks as Kernel Learners: The Silent Alignment Effect — While the loss is flat, student singular vectors align with teacher singular vectors, and this can be detected with the NTK
- Are Emergent Abilities of Large Language Models a Mirage? — Emergence is in the eye of the metric used to measure it
- Training Compute-Optimal Large Language Models — Chinchilla scaling law paper
- Emergent Abilities of Large Language Models — Shows that novel capabilities emerge with more compute

Theoretical approaches to emergence

- Grokking as the Transition from Lazy to Rich Training Dynamics
- Grokking as a First Order Phase Transition in Two Layer Networks, Rubin et al.
- Blake Bordelon - Infinite limits and scaling laws of neural networks - IPAM at UCLA
- A Theory for Emergence of Complex Skills in Language Models
- On neural scaling and the quanta hypothesis
- Lecture Notes on Infinite-Width Limits of Neural Networks; Cengiz Pehlevan and Blake Bordelon

- Disordered Dynamics in High Dimensions: Connections to Random Matrices and Machine Learning; Blake Bordelon, Cengiz Pehlevan
- Applications of Statistical Field Theory in Deep Learning; Zohar Ringel et al.
- Statistical Field Theory for Neural Networks, Moritz Helias, David Dahmen
- Lecture notes: From Gaussian processes to feature learning, Moritz Helias et al

References

[AMG24] Achour, Malgouyres, and Gerchinovitz. The loss landscape of deep linear neural networks: a second-order analysis. *Journal of Machine Learning Research (JMLR)*, 2024. URL: https://arxiv.org/abs/2107.13289.

[SMG14] Saxe, McClelland, and Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations (ICLR)*, 2014.

[TAJ24] Tu, Aranguri, and Jacot. Mixed dynamics in linear networks: Unifying the lazy and active regimes. 2024. URL: https://arxiv.org/abs/2405.17580.